

SELENE

An On-Device, Sensor-Fused AI Companion for the iQOO 15

“When connectivity disappears, SELENE’s intelligence doesn’t.”

THE PROBLEM

1. Modern smartphones contain numerous sensors, but their signals are typically used by separate, isolated features.
2. Connectivity gap: Cloud-dependent assistants lose functionality in highways, ghat roads, tunnels, basements, crowded transit and disaster zones.
3. Context gap: Individual sensors can detect motion, location or sound, but rarely combine them to understand what is actually happening.
4. Users are still expected to initiate actions manually, even when the phone already has enough contextual information to assist.
5. The result is a device that senses a great deal and understands very little.
6. What is missing is not more sensors, but a layer that connects them.

THE CORE IDEA

1. Transforms the iQOO 15 into a continuously aware, context-driven AI companion.
2. Fuses motion, location, vision, sound, connectivity and battery into one situational model.
3. Understands context proactively, instead of waiting for explicit commands.
4. Activates only the intelligence required for the current situation, reducing unnecessary computation.
5. Operates through the core loop: Sense, Fuse, Understand, Reason, Act, Learn.
6. A multi-model, fully on-device AI stack reasons over that context without continuous cloud access.
7. Result: the phone can remain situationally intelligent even when the network cannot.

SELENE’S INTELLIGENCE PIPELINE

1. **Sense** — Collect relevant sensor, device and environmental signals.
2. **Fuse** — Synchronize independent signals to reduce isolated-sensor errors.
3. **Understand** — Convert raw measurements into meaningful activities, events and context.
4. **Reason** — Select the appropriate AI model or capability for the situation.
5. **Act** — Assist, alert, protect, communicate or automate the required response.
6. **Learn** — Improve contextual personalization locally, with user permission.
7. Sensing, reasoning and action stay connected instead of operating as separate features.

SENSOR INTELLIGENCE

1. Combines motion, orientation, location, light, proximity, sound, vision, connectivity and battery state.

2. Identifies events through relationships between signals, not isolated thresholds.
3. Builds a compact representation of activity, motion state, location familiarity, ambient conditions, connectivity, battery state and time.
4. Converts raw sensor streams into structured context the AI can reason over, and uses it to determine which capability should activate.
5. Keeps contextual processing local instead of continuously sending raw sensor information to the cloud.

MULTI-MODEL ON-DEVICE AI

1. Uses specialized models for specialized tasks, avoiding unnecessary use of a large model.
2. **Safety Detectors** — Detect patterns associated with crashes, falls, snatches and relevant acoustic events.
3. **Qwen3-1.7B** — Handles lightweight intent recognition, commands, routing and simple reasoning.
4. **Qwen3-4B** — Handles deeper reasoning, emergency summaries, first-aid-oriented guidance and complex offline assistance.
5. **Qwen3-VL** — Provides visual understanding of signs, landmarks, injuries, documents and other visual context.
6. **Whisper** — Offline speech-to-text. **Piper** — Offline text-to-speech. **bge-small** — Local semantic retrieval from trusted offline information.
7. Larger models are loaded only when required and unloaded once their tasks complete, balancing intelligence, latency, battery and memory.
8. Safety architecture follows the principle: deterministic systems detect; AI models reason, explain and assist.
9. An LLM is never the sole trigger for a safety-critical action.

GUARDIAN INTELLIGENCE

1. Separates fast deterministic event detection from slower generative AI reasoning, improving predictability and responsiveness.
2. Detects probable crashes using multi-sensor event correlation: speed, acceleration, impact, orientation, acoustic events and post-impact inactivity.
3. Initiates a cancellable SOS countdown once multiple signals agree and confidence is sufficient.
4. Generates a structured emergency packet containing location, event type and available contextual information.
5. Transmits the packet when communication is available, and queues it when connectivity is not.
6. Converts the phone from a passive emergency contact tool into an active incident-detection layer.

FALSE-ALARM PREVENTION

1. Multi-signal confirmation, not a single sensor anomaly, raises event confidence.
2. Example crash signature: Speed + impact + orientation + acoustic event + inactivity.

3. Example fall signature: Free-fall + impact + inactivity.
4. Example snatch signature: Proximity change + anomalous motion + wearable disconnection.
5. This distinguishes significant events from ordinary movements such as dropping the phone, sitting down or normal handling.

ANTI-THEFT INTELLIGENCE

1. Learns normal device-handling patterns locally.
2. Identifies deviations using unexpected motion, proximity changes, handling behaviour and wearable disconnection.
3. A probable snatch triggers rapid device lock, authentication, alarm, camera capture where supported and location preparation.
4. Preserves contextual evidence rather than treating theft protection as only a screen-lock function, with vision preparing a description for trusted contacts.

OFFLINE EMERGENCY BRAIN

1. Provides emergency assistance without cloud dependency, combining bge-small, a trusted first-aid corpus, Qwen3-4B and Piper.
2. Supports first-aid guidance, emergency summaries, voice instructions, siren and strobe activation and queued SOS information.
3. Uses local retrieval to ground responses in a trusted offline information base.
4. Voice output allows assistance when the user cannot conveniently interact with the screen.
5. Visual injury analysis provides first-aid guidance, not medical diagnosis.

VISUAL INTELLIGENCE

1. Turns the camera into a situational context sensor, not merely an image-capture device.
2. Qwen3-VL can interpret road signs, landmarks, injuries, documents, receipts and whiteboards.
3. Combines visual information with GNSS context, supplying location clues when GNSS alone is insufficient.
4. Enriches emergency reports with environmental information visible to the camera.
5. Extends the phone's understanding from "where am I?" to "what is around me?".

CONNECTIVITY AND BATTERY INTELLIGENCE

1. Treats connectivity and remaining power as dynamic inputs to decision-making, not simply on/off features.
2. Monitors cellular, Wi-Fi, Bluetooth, GNSS and battery state.
3. When connectivity disappears, SELENE shifts toward local reasoning and queues important information for later transmission, creating a store-and-forward architecture.
4. Under critical battery conditions, Survival Mode prioritizes safety, emergency functions and essential communication.
5. Suspends non-essential computation to extend the availability of critical functions.

6. Intelligence therefore adapts not only to what is happening, but also to what resources remain available.

VOICE-FIRST INTELLIGENCE

1. Enables a complete offline voice interaction loop: Whisper → Router → Reasoner → Piper.
2. Whisper converts speech into text locally; the Router identifies intent and selects the appropriate capability.
3. The Reasoner handles deeper contextual processing; Piper converts the response into speech.
4. Enables natural interaction when the screen is inconvenient, occupied or unsafe to use.

CONTEXT-AWARE AUTOMATION AND PERSONAL INTELLIGENCE

1. Moves the phone from manual commands toward contextual assistance.
2. Can adapt focus modes, notifications, reminders, routines, IR control and voice actions.
3. Learns user-specific patterns locally and with explicit consent: familiar places, normal routines, common activities and typical device handling.
4. Uses learned patterns to identify meaningful deviations, providing a baseline for what is normal.
5. Personalization remains local instead of requiring continuous transmission of behavioural information to the cloud.

THE SPECTRUM OF USE

The same sensors, models and pipeline serve three different levels of need. Everyday use establishes value, professional use establishes trust, and critical use establishes purpose. Because the intelligence layer is shared, the retrieval engine that searches a student's notes is the same one that searches the first-aid corpus, and the vision model that reads a receipt is the same one that reads a road marker after a crash.

Everyday

1. Wakes the user within a light-sleep window, followed by a spoken daily brief.
2. Delivers a departure brief on leaving home: schedule, cached weather, priority messages and a belongings check based on Bluetooth presence and learned routine.
3. Adapts focus modes automatically, silencing the phone in class or during known commitments.
4. Converts lectures, discussions and whiteboard photographs into structured notes, summaries and flashcards, entirely on-device.
5. Uses the camera to explain textbook problems, medicine labels, food packaging and signboards in other languages.
6. Controls fans, air conditioners and televisions by voice through the IR blaster, with no internet required.

Professional

1. Transcribes meetings locally and extracts action items, keeping confidential discussion on the device.
2. Reads photographed contracts, summarising clauses and flagging deadlines, amounts and obligations.
3. Captures receipts and visiting cards into logged expenses and contacts, and triages email, summarising only what is relevant.
4. Recognises travel context, shifting to voice-first interaction while arming incident detection.
5. Operates in basements, tunnels and field sites where cloud assistants become unavailable.

Critical

1. Detects probable falls, crashes and theft through multi-signal agreement rather than single thresholds.
2. Initiates a cancellable countdown before any emergency response.
3. Prepares a structured emergency packet with location, event type and, where available, visual landmarks.
4. Delivers offline first-aid guidance by voice when the screen cannot be used.
5. Activates siren and strobe output to attract assistance in the immediate environment.
6. Enters Survival Mode under critical battery conditions, transmitting last known location while power remains.

ONE DAY WITH SELENE

Imagine one phone accompanying different people through ordinary and extraordinary moments.

Morning — Meena

1. A 68-year-old user falls in the bathroom.
2. SELENE detects free-fall, impact, orientation change and prolonged inactivity.
3. If connectivity exists, the configured emergency contact can be alerted.
4. If connectivity is unavailable, the event is queued locally while a local alarm can assist.

Afternoon — A Student

1. The phone records the lecture locally, without requiring cloud processing.
2. Whisper transcribes it offline, and Qwen3-VL understands the board and other visual material.
3. SELENE converts the combined information into structured notes and flashcards.
4. Classroom audio and content can remain on-device.

Evening — A Commuter

1. A sudden proximity change, anomalous movement and wearable disconnection create a probable snatch signature.
2. SELENE correlates these signals instead of relying on one threshold.
3. The phone can lock immediately and activate its configured security response.

Emergency — A Rider

1. A crash occurs on a remote ghat road.
2. SELENE combines multiple signals to confirm the event instead of relying on a single sensor.
3. It prepares an SOS and structured emergency packet.
4. Available visual information can identify road markers or landmarks and enrich the situation description.
5. If there is no network, the packet is stored locally for later transmission.
6. Offline first-aid-oriented guidance can be delivered through voice.
7. Different people. Different situations. One intelligent companion.

PRIVACY BY DESIGN

1. Built around local AI inference wherever possible.
2. Avoids continuous cloud uploading of raw sensor streams.
3. Acoustic intelligence can rely on event classification rather than continuous audio storage.
4. Personal routines, documents, meetings, classroom content and notes are processed and stored on-device.
5. Personalization remains opt-in and locally controlled.
6. Privacy is therefore part of the architecture rather than an additional feature.

PERFORMANCE TARGETS

1. Designed around Snapdragon and Hexagon NPU acceleration through Qualcomm AI Hub.
2. Targeted performance: approximately 22 tokens per second for Qwen3-4B; approximately 2.5 seconds for specialist model loading; approximately 8 seconds from impact detection to SOS-ready state in a fully offline scenario.
3. Selective model activation and resource-aware execution maintain responsiveness while controlling battery and memory use.

WHAT MAKES SELENE DIFFERENT

1. SELENE is not simply an AI chatbot, safety app or sensor monitor.
2. It combines Sensor Fusion, On-Device AI and Offline Intelligence in one layer connecting safety, vision, voice, connectivity, battery awareness, automation and personal context.
3. Converts isolated sensor readings into situational understanding.
4. Separates safety-critical detection from generative AI reasoning.
5. Makes connectivity, battery and physical surroundings part of AI decision-making.
6. Reuses existing smartphone sensing to create a unified context-aware system, adapting to the situation and to available resources.
7. The key shift is from feature-based sensing to situation-based intelligence.

FINAL VISION

1. A phone should not become unintelligent when the network disappears.

2. Sensors should not operate as isolated features when their combined signals can reveal context.
 3. AI should activate when needed instead of consuming maximum resources continuously, and users should not have to configure the phone for every situation.
 4. Intelligence should understand what is happening, not merely respond to what was asked.
 5. SELENE transforms the smartphone from a device that waits for commands into a companion that understands context.
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Sense. Fuse. Understand. Reason. Act. Learn.

Turn disconnected moments into intelligent zones.

SELENE — When connectivity disappears, intelligence doesn't.